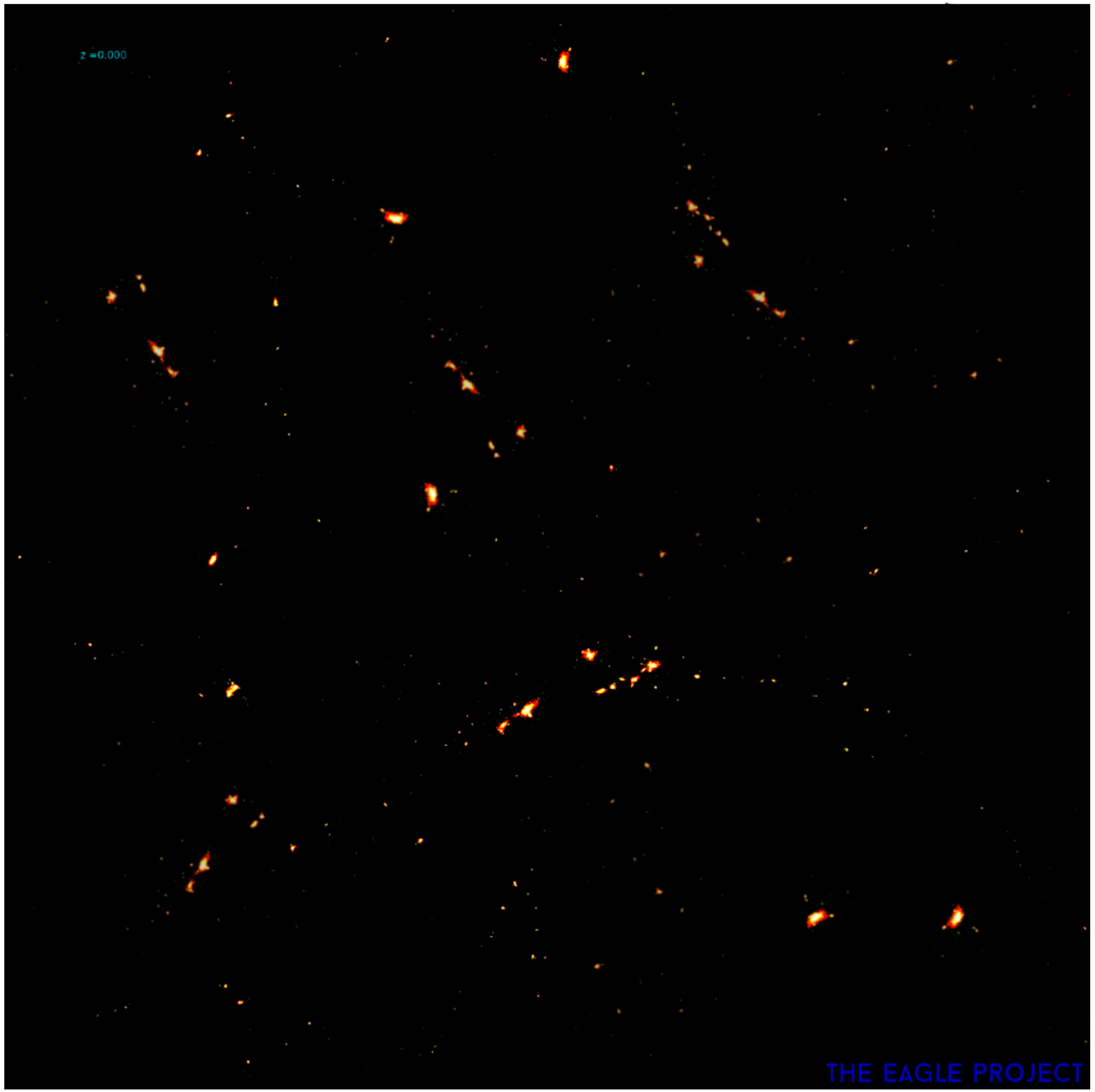
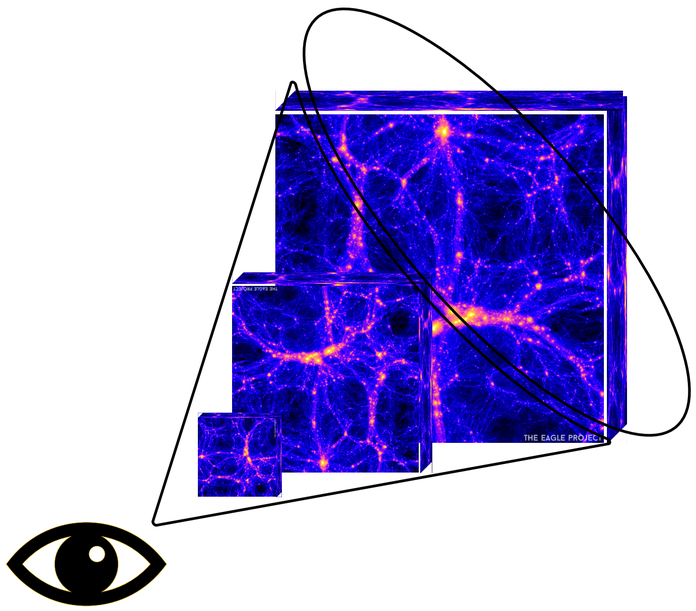
… and we see a lightcone



$100\ h^{-1}$ Mpc

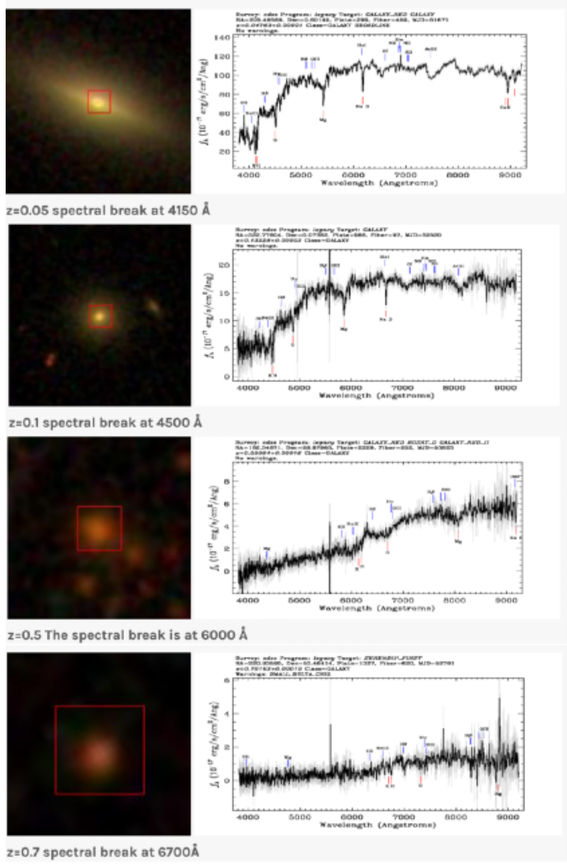


→ We need 3D mapping of galaxy positions

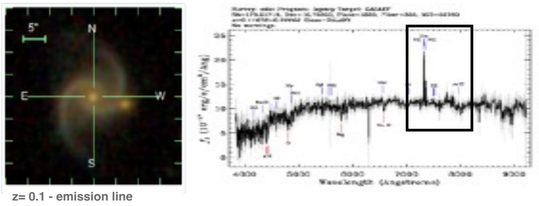
i.e **need to estimate galaxy distances**

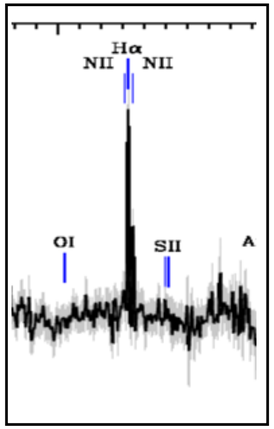
Estimate of galaxy distances

- Measure the redshift z of the galaxies, which to first order is a distance estimate



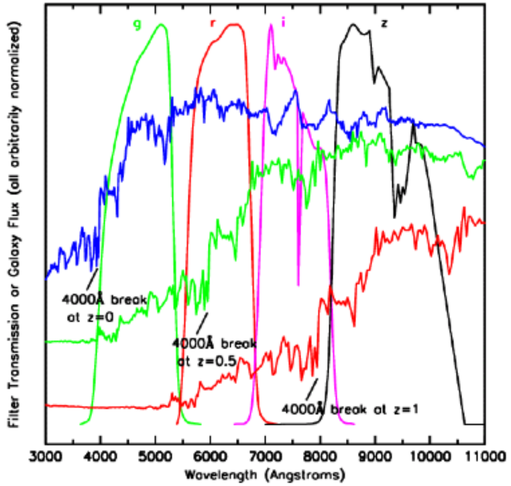
As observed flux decreases with redshift, getting a measurement get increasingly harder

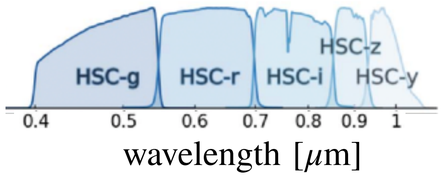


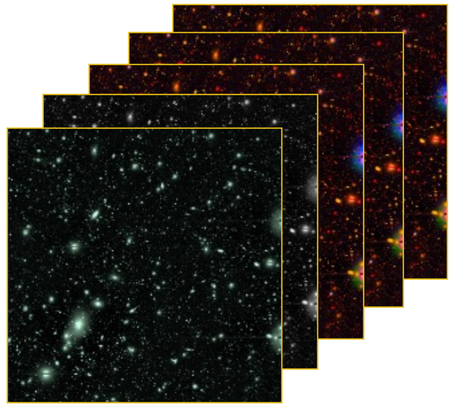


Emission line galaxies can be detected at higher redshift

→ *Euclid*







Multiband photometric observations

Very low resolution, integrated spectrum.

Important uncertainty on redshift (5-10 %)